

Disaster Damage Assessment Using Semantic Change Detection and Deep Learning: An Automated Framework for Damage Classification

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Abstract:

Timely and accurate disaster damage assessment is essential for effective emergency response and resource allocation. This project proposes an automated framework that combines Siamese Neural Networks with YOLOv8-based instance segmentation to analyze before-and-after imagery. By integrating change detection algorithms with object recognition, the framework accurately identifies structural footprints and quantifies damage severity based on pixel-level change percentage. The system produces GIS-compatible damage classification reports, providing emergency management teams with timely, actionable information to support informed decision-making and effective disaster response planning.

Keywords:

Convolutional Neural Networks (CNNs), Siamese U-Net, YOLOv8 Segmentation, Satellite Imagery, Drone Imagery, xBD Dataset, xView2 Dataset, Building Collapse Detection.

Introduction:

Natural disasters cause billions of dollars in infrastructure losses each year, making rapid and accurate damage assessment essential for effective emergency response and resource allocation. For example, the 2020 Atlantic hurricane season alone resulted in more than \$40 billion in damages, underscoring the urgent need for improved post-disaster assessment methodologies.

Traditional approaches, such as manual field surveys and 2D satellite image interpretation, are timeconsuming, often requiring weeks to complete, and frequently provide incomplete or inconsistent results. These limitations are driven by restricted access to disaster zones, subjective human judgment, and the lack of automated quantification methods.

Recent advances in deep learning offer new opportunities to overcome these challenges.

Semantic change detection methods, combined with object segmentation, enable automated identification and quantification of structural damage directly from before-and-after imagery. Convolutional neural networks (CNNs), particularly Siamese architectures, can detect pixel-level changes, while YOLOv8 segmentation provides precise building footprints.

This research proposes an automated framework that integrates Siamese U-Net–based change detection with YOLOv8 segmentation to process satellite and drone imagery. The framework outputs GIS-compatible damage classification reports, categorizing structures into Minor, Major, or Destroyed based on pixel-level change percentages. By reducing assessment time from weeks to minutes and improving accuracy and consistency, the system has the potential to significantly enhance disaster response and recovery planning

Literature Review:

[1] S. A. Chen et al., Benchmark dataset for automatic damaged building detection from posthurricane remotely sensed imagery (2018)

This paper introduces a benchmark dataset specifically designed for damaged building detection after hurricanes. The dataset provides annotated post-disaster imagery, enabling researchers to train and evaluate deep learning models for automatic damage classification. The study highlights the importance of standardized datasets for improving reproducibility and advancing disaster response applications. Its contribution laid the groundwork for subsequent large-scale datasets such as xBD and xView2.

[2] I. Demir et al., DeepGlobe 2018: A challenge to parse the earth through satellite images (2018)

The DeepGlobe challenge focused on parsing satellite imagery for tasks such as road extraction, land cover classification, and building detection. By organizing a global competition, the authors provided a benchmark for evaluating computer vision models on geospatial tasks. The challenge demonstrated the effectiveness of convolutional neural networks (CNNs) in extracting semantic information from satellite imagery, which is directly applicable to disaster damage assessment workflows.

[3] A. Fujita et al., Damage detection from aerial images via convolutional neural networks (2017)

This study applied CNNs to aerial imagery for detecting building damage after disasters. The authors demonstrated that deep learning models outperform traditional image processing techniques in identifying collapsed or severely damaged structures. Their work validated the feasibility of CNN-based approaches for rapid disaster assessment and inspired further research into integrating segmentation models like U-Net and YOLO for more precise footprint detection.

[4] R. Gupta et al., Creating xBD: A dataset for assessing building damage from satellite imagery (2019)

The xBD dataset is the largest open-source benchmark for building damage assessment, covering multiple disaster types such as earthquakes, floods, and hurricanes. It provides pre- and post-disaster satellite imagery with labeled damage categories (No Damage, Minor, Major, Destroyed). This dataset has become the foundation for training and evaluating deep learning models in disaster response research. Its geo-referenced annotations make it directly compatible with GIS platforms, aligning perfectly with automated damage classification systems.

IMPLEMENTATION

Overall System Architecture

The proposed disaster assessment framework is designed as a unified multi-stage pipeline that processes satellite and drone imagery through deep learning models. The system begins with data collection, followed by preprocessing, semantic change detection, object segmentation, and GIS-compatible damage classification. Each module operates collaboratively, allowing the system to handle diverse disaster scenarios such as earthquakes, floods, cyclones, and building collapses. This architecture ensures rapid, scalable, and accurate analysis suitable for real-world emergency response.

Data Sources and Acquisition

The implementation uses reliable geospatial data sources obtained from satellite APIs and drone surveys. Satellite imagery can be fetched from platforms such as Copernicus EMS, Sentinel-2, and Planet Labs, while benchmark datasets such as **xBD** and **xView2** are incorporated during training and evaluation. These datasets provide pre- and post-disaster imagery with labeled building damage categories, ensuring robustness and generalization of the deep learning models.

Preprocessing Workflow

Before feeding the images into the models, the system performs standard preprocessing operations:

- Resizing image pairs to a fixed resolution (512×512).
- Normalizing pixel values using mean and standard deviation vectors.
- Aligning pre- and post-disaster images to ensure spatial consistency.

This preprocessing ensures that the Siamese U-Net receives properly formatted tensors for change detection.

Deep Learning Model Training

Damage detection is performed using a **Siamese U-Net** architecture. Two identical encoder branches process the pre- and post-disaster images, extracting feature maps that are fused and decoded into a binary change mask.

Building segmentation is performed using **YOLOv8-Segmentation**, which detects and segments building footprints in the post-disaster image.

The training process uses benchmark datasets (xBD, xView2) with augmentation techniques such as rotation, scaling, and brightness adjustment to improve robustness. The models are implemented in **PyTorch** and integrated into a Flask-based web application for real-time inference.

Damage Severity Classification

The system calculates damage severity using the **Intersection over Area (IoA)** method:

- Changed pixels (from Siamese U-Net) are intersected with building masks (from YOLOv8).
- The ratio of changed pixels to total building pixels gives the **damage percentage**.
- Classification thresholds:
 - **Destroyed:** $> 70\%$
 - **Major:** $25\text{--}70\%$
 - **Minor:** $5\text{--}25\%$
 - **Safe:** $\leq 5\%$

This classification is displayed visually with color-coded bounding boxes (Green = Safe, Orange = Major, Red = Destroyed) and summarized in a GIS-compatible report panel.

Tools and Technologies Used • Python 3.8

- **PyTorch** for Siamese U-Net training and inference
- **Ultralytics YOLOv8** for building segmentation
- **OpenCV, NumPy** for image preprocessing and visualization
- **Flask** for web deployment and interactive analysis
- **GDAL, Rasterio, GeoPandas** for GIS compatibility and spatial data handling

Results

The developed system demonstrates effective performance in post-disaster semantic change detection and building-level damage classification. A Siamese U-Net architecture is employed to identify change regions between pre- and post-event imagery, while YOLOv8 segmentation is used to extract building footprints and enable object-level analysis.

The system categorizes structural damage into four levels: Destroyed, Major, Minor, and Safe, providing an interpretable and operationally meaningful output for emergency response applications. The final output comprises a GIS-compatible visual assessment, including:

- Red change-detection overlays highlighting affected areas,
- Color-coded bounding boxes around individual buildings,
- Damage labels indicating categorical severity, and
- A summary panel presenting the total number of detected structures and the count of buildings in each damage category.

The categorical nature of the output ensures clarity, rapid interpretation, and effective utility for field teams, particularly in time-critical disaster response scenarios.

Conclusion

The initial project plan envisioned a more detailed analytical framework, including explicit damage percentage visualization and volumetric debris estimation derived from 3D reconstruction. In practice, the implemented system strategically focuses on providing categorical damage classification only—Destroyed, Major, Minor, and Safe.

This design pivot proved highly advantageous. By prioritizing the categorical output, the system significantly reduces visual complexity and enhances usability for emergency responders who require immediate and unambiguous situational information. Although the primary output is categorical, the underlying system is highly quantitative; the precise damage percentage is internally calculated (via the IoA method) and displayed in the summary statistics, ensuring quantitative integrity while maintaining operational simplicity.

Thus, although the system does not currently incorporate volumetric analysis or 3D modeling, the resulting workflow is faster, more stable, and operationally more practical than the originally proposed design. The approach maintains full GIS compatibility, with each building represented by its geographic location and its corresponding damage category and percentage.

Future Scope

Future enhancements to the system can be explored across several technical dimensions. First, integrating damage-percentage visualization directly into bounding-box labels would enable quantitative interpretation alongside categorical damage classes. Advanced transformer-based architectures such as SegFormer or Swin-UNet may further enhance contextual reasoning and segmentation quality. Leveraging geospatial foundation models—including Prithvi and SatMAE—represents a promising direction for improving generalization across varied disaster scenarios and geographic domains.

To increase robustness under challenging imaging conditions, incorporating SAR datasets (e.g., Sentinel-1) can provide reliable damage assessment during cloud cover, smoke, or nighttime scenarios. Likewise, integrating LiDAR data or adopting lightweight 3D reconstruction techniques could support future work on volumetric debris estimation. Edge-optimized deployment on platforms such as NVIDIA Jetson would enable real-time, on-site analysis for field teams.

In addition, developing automated reporting pipelines capable of generating structured GIS layers, dashboards, and analytical summaries can streamline operational workflows. Finally, coupling observed damage patterns with predictive hazard and vulnerability models opens a pathway toward early-warning and proactive disaster-response intelligence.

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